

ZAIN UL ABIDIN

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[Google Scholar](#) | [ORCID](#) | [GitHub](#) | [LinkedIn](#) | [ResearchGate](#) | [Portfolio](#)

RESEARCH PROFILE

Final-year Computer Systems Engineering student (CGPA 3.79/4.00, PEC/Washington Accord accredited) with a strong undergraduate research record in machine learning, computer vision, and image processing. Co-author of six papers: one published, one accepted to IEEE Xplore, three under review in Q1 journals by Taylor and Francis and Elsevier (impact factors up to 4.9), and one in preparation. Research interests span deep learning, computer vision, image and signal processing, and their application to real-world engineering problems. Seeking a fully funded Master's or PhD position in machine learning, computer vision, or a related field.

RESEARCH INTERESTS

Machine Learning and Deep Learning for Industrial Applications | Computer Vision and Image Processing | Non-Destructive Testing and Evaluation | Digital Signal Processing | Structural Health Monitoring

EDUCATION

B.Eng. in Computer Systems Engineering

Sep 2022 - May 2026

Sukkur IBA University, Sukkur, Sindh, Pakistan | [iba-suk.edu.pk](#)

- CGPA: 3.79 / 4.00 (A = 4.0, US 4.0 GPA scale; all courses taught in English)
- PEC Accredited, Level-II, Washington Accord Substantial Equivalence
- Relevant courses: Artificial Intelligence and Machine Learning (A), Digital Signal Processing (A), Probability and Statistics (A), Signals and Systems (A), Complex Variables and Transforms (A), Digital System Design (A)

PUBLICATIONS

Published

1. Mitigation of DDoS Attacks with Machine Learning, Deep Learning, and Transformers [\[First Author\]](#)

Zain Ul Abidin, Haseeb, Muhammad Younis, Habeeban Memon, Junaid Ahmed Bhatti

International Journal of Multidisciplinary Conference Proceedings (IJMCP), Vol. 2(1), 2024. DOI: 10.61503/Ijmcpc.v2i1.181

Benchmarked ML and DL models for multi-class DDoS detection on CICDDoS2019. LightGBM and XGBoost achieved the best accuracy-speed balance for real-time network security.

Accepted (In Press)

2. Debond Detection in CFRP Structures Using Optical Pulse Thermography and Deep Learning [\[Second Author\]](#)

Habeeban Memon, Zain Ul Abidin, Junaid Ahmed, Ayesha Siddiqua

iCoMET 2025, IEEE Xplore, Scopus (EI Compendex)

Compared YOLOv8, Faster R-CNN, and Mask R-CNN for CFRP debond detection. Mask R-CNN achieved the highest accuracy with pixel-level defect segmentation.

Under Review

3. Metal Defect Detection in Pulsed Thermographic Imaging via Temporal Eigen Projection and Robust Low-Rank Tensor Completion [\[First Author\]](#)

Zain Ul Abidin, Junaid Ahmed, Dong Wang, Guiyun Tian

Nondestructive Testing and Evaluation, Taylor and Francis | IF 4.2 | Q1 | WoS and Scopus

Proposed Eigen-RHOTC combining temporal eigen projection with robust higher-order tensor completion to isolate ECPT defect features. Outperforms prior methods in SNR and processing speed.

4. Efficient Defect Detection in Metals via Keyframe Extraction and Accelerated Sparse Decomposition [\[Second Author\]](#)

Junaid Ahmed, Zain Ul Abidin, Guiyun Tian, Dong Wang

Quantitative InfraRed Thermography Journal, Taylor and Francis | IF 4.9 | Q1 | WoS and Scopus

Entropy- and density-based keyframe selection reduces thermal data volume. Accelerated CUR decomposition extracts weak defect signals from noisy backgrounds.

5. Improving Image Super-Resolution: A Hybrid Approach Using DT-CWT and Transformer Networks [Fifth Author]

Pan Zhiwen, Faisal Sahito, Aakash Kumar, Junaid Ahmed, Zain Ul Abidin

Image and Vision Computing, Elsevier | IF 4.2 | CiteScore 7.1 | Q1 | WoS and Scopus

A DT-CWT and Transformer hybrid for image super-resolution that preserves fine structural detail better than current state-of-the-art baselines.

In Preparation

6. Automated Defect Detection in CFRP Composites Using Multi-Modal Thermal Representations and Deep Learning [First Author]

Zain Ul Abidin, Habeeban Memon, Junaid Ahmed, Ayesha Siddiqua | Target journal to be confirmed

A three-channel thermal tensor (raw intensity, PCA variance, TSR temporal dynamics) improved segmentation accuracy by up to 154% over single-channel baselines, with physics-informed depth estimation for subsurface defects.

RESEARCH AND TECHNICAL PROJECTS

OPT Defect Detection, Segmentation and Depth Estimation Framework

2025 - Present

NDT/NDE | CFRP Composites | Deep Learning | Related to Publications 2, 3 and 6

- End-to-end pipeline for automated detection, segmentation, and physics-informed depth estimation of subsurface defects in CFRP composites from Optical Pulse Thermography image sequences. Integrates YOLOv8, semantic segmentation, and ML regression.

Repository: github.com/zainulabidinrain/OPT_Defect_Inspection_Framework

Real-Time Speech Emotion Recognition via LSTM

2025

Speech Processing | LSTM | Toronto Emotional Speech Set (TESS)

- LSTM classifier trained on TESS with noise injection and time-shifting augmentation; achieves real-time inference on live audio streams.

Repository: github.com/zainulabidinrain/speech-emotion-recognition-lstm

Hand Gesture Recognition and Voice Conversion

2024

Computer Vision | CNN | SVM | Accessibility

- CNN and SVM pipeline for sign language gesture classification; gestures converted to synthesised speech in real time for users with hearing or speech impairments.

Repository: github.com/zainulabidinrain/hand-gesture-voice-conversion

Gold Price Forecasting Dashboard

2025

Time-Series Forecasting | Ensemble Learning | Streamlit

- Ensemble of XGBoost, LSTM, and Random Forest models trained on historical market data, visualised through an interactive Streamlit dashboard with side-by-side model comparison.

Repository: github.com/zainulabidinrain/gold-price-prediction

Speech Enhancement via Spectral Subtraction and Wiener Filtering

2025

Signal Processing | Python

- Real-time pipeline estimating background noise from degraded speech and applying Wiener filtering to recover intelligibility.

Repository: github.com/zainulabidinrain/speech-enhancement-wiener-filter

Image Compression Analysis: JPEG vs JPEG2000

2024

Image Processing | DCT | DWT

- DCT-based JPEG and DWT-based JPEG2000 implemented from first principles; quality benchmarked via PSNR, SSIM, and compression ratio.

Repository: github.com/zainulabidinrain/image-compression-analysis

AI-Based Android Controlled Vision Robot

2024

Edge AI | ESP32-CAM | TensorFlow Lite

- ESP32-CAM with TensorFlow Lite for on-device real-time object detection; demonstrates edge AI deployment on low-cost embedded hardware.

Repository: github.com/zainulabidinrain/ai-vision-robot-esp32

WORK EXPERIENCE

Artificial Intelligence Intern

Jul 2025 - Aug 2025

Mirai School of Technology | Remote

- Designed and deployed a Conversational Order Assistant using agentic AI workflows (n8n, Gemini AI), automating order capture, validation, logging, and email confirmation via RESTful APIs.

Software Engineering Intern

Jun 2025 - Jul 2025

Sukkur Electric Power Company (SEPCO) | On-site, Sukkur, Sindh

- Built the Mute Meter Monitoring Dashboard (Python, Streamlit, SQLite) with GIS zone filters and bulk import/export, reducing manual field inspection workload by an estimated 40%.

TECHNICAL SKILLS

Programming: Python, MATLAB, SQL, Verilog

ML and DL Frameworks: TensorFlow, PyTorch, Keras, scikit-learn, XGBoost, LightGBM

Computer Vision: OpenCV, YOLOv5/v8, Mask R-CNN, Faster R-CNN, PyWavelets

Scientific Computing: NumPy, SciPy, Pandas, Matplotlib, Seaborn

Tools and Platforms: Git, GitHub, Jupyter Notebook, Streamlit, MySQL, SQLite, n8n

Embedded Systems: ESP32, Arduino (proficient) | Raspberry Pi (basic)

HONOURS AND AWARDS

Prime Minister's Laptop Award

2024

National merit-based award for academic excellence, Government of Pakistan.

Sindh Educational Endowment Fund (SEEF) Scholarship

2022 - Present

Fully merit-based scholarship covering undergraduate tuition fees for the full duration of the degree.

Top Academic Ranking, Computer Systems Engineering

2022 - Present

Consistently ranked among the highest-performing students in the cohort; CGPA 3.79 / 4.00.

SELECTED CERTIFICATIONS

- Machine Learning Specialization, DeepLearning.AI / Coursera
- Mathematics for Machine Learning and Data Science Specialization, DeepLearning.AI / Coursera
- Deep Learning Specialization: Advanced AI, Hands-on Lab, Udemy
- Large Language Models, Udemy
- Google Prompting Essentials Specialization, Google / Coursera
- Google IT Support Specialization, Google / Coursera
- Career Essentials in GitHub Professional Certificate, GitHub / LinkedIn Learning
- SQL for Developers and Data Analysts (MySQL), Udemy

LANGUAGES

English (Professional) | **Urdu** (Native) | **Sindhi** (Native) | **Punjabi** (Native)

REFERENCES

Academic and professional references are available on request. Research supervisors for Publications 3, 4, and 6 can speak directly to research contributions.